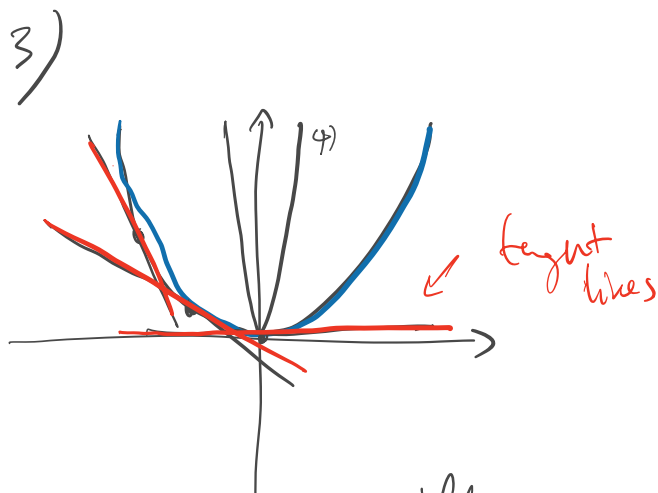
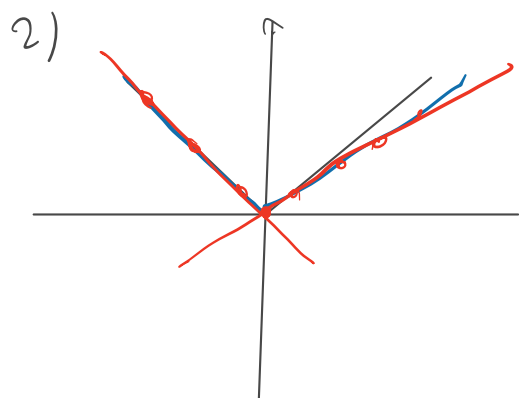
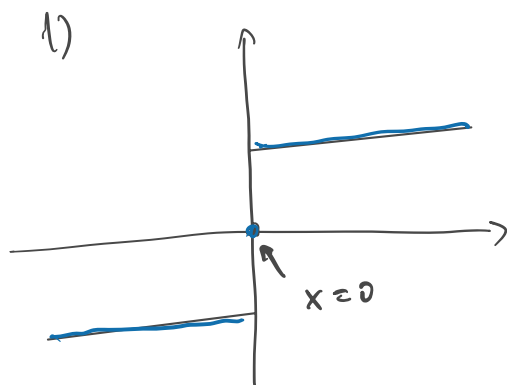


Lecture 8:



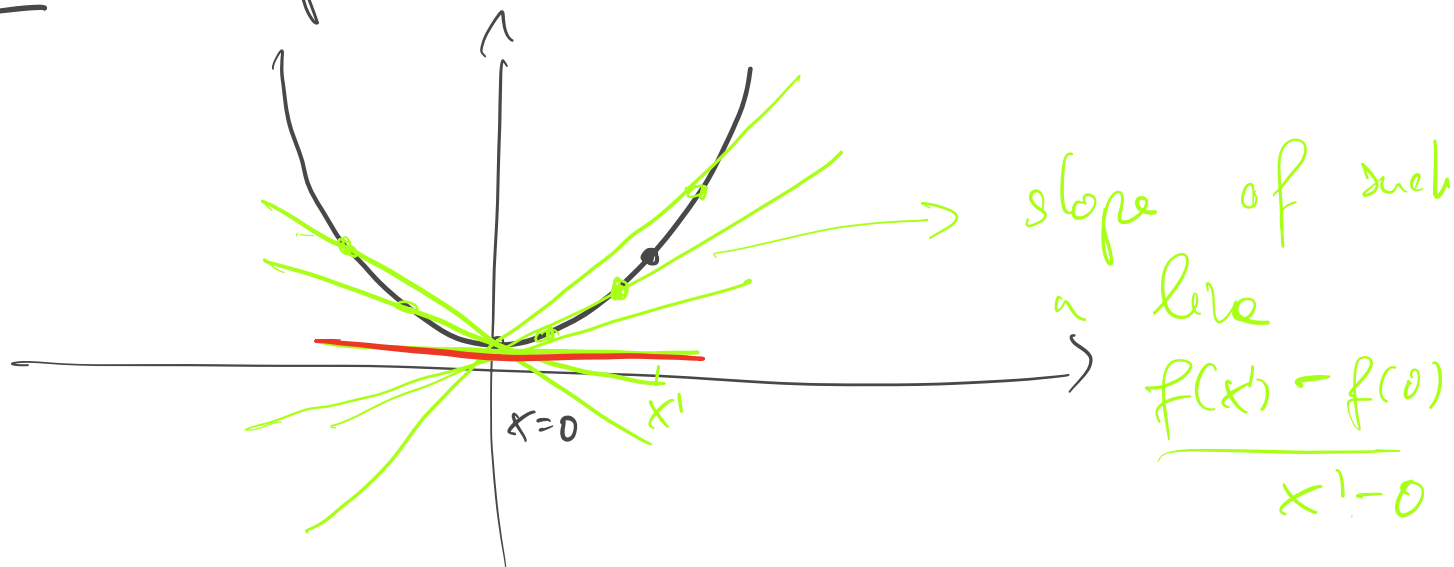
Q: What are the differences in these graphs.

1) Not cont. at $x=0$!

2) is not diff'able at $x=0$!

Goal: Formalise this property (of function) of having a well-defined tangent.

How to compute this slope/tangent here?



Defn Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a function. $x_0 \in \mathbb{R}$, $a \in \mathbb{R}$.

We say $f(x)$ converges to a as $x \rightarrow x_0$ if

$\forall \epsilon > 0 \exists \delta > 0$ st. $\forall x \neq x_0$ $|x - x_0| < \delta \Rightarrow |f(x) - a| < \epsilon$.

Rem. f is cont. at $x_0 \in \mathbb{R} \Leftrightarrow f(x)$ converges to $f(x_0)$
as $x \rightarrow x_0$.

Notation: $\lim_{x \rightarrow x_0} f(x) = a$.

Defn. $f: \mathbb{R} \rightarrow \mathbb{R}$, $x_0 \in \mathbb{R}$, $f' \in \mathbb{R}$.

We say f is differentiable at x_0 with derivative f'

if $\lim_{x \rightarrow x_0} \frac{f(x) - f(x_0)}{x - x_0} = f'(x)$

• The derivative $f': \mathbb{R} \rightarrow \mathbb{R}$ of f

is defined as

$$f'(x) = \begin{cases} \lim_{x' \rightarrow x} \frac{f(x') - f(x)}{(x' - x)} & \text{if it exists} \\ 0 & \text{else.} \end{cases}$$

What do we need to define
 f' of f ?

- 1) Define the above notion of conv.
"tend to" for functions
- 2) Differentiability at a point. dep. on the limit
- 3) Global def of diff'ble.
∃ a function that computes the above limit
- 4) Define the derivative, the above limit

→ involves a choice.

Observations :

$$1) f(x) = c, \text{ then } f'(x) = 0$$
$$\left(\frac{f(x') - f(x)}{x' - x} = \frac{c - c}{x' - x} = 0 \right)$$

$$2) f(x) = ax + b, \text{ then } f' = a$$

$$\left(\frac{f(x') - f(x)}{x' - x} = \frac{a(x' - x)}{(x' - x)} = a \right)$$

3) If f is differentiable then
 f is continuous.

Pf. Pick $x \in \mathbb{R}$

$$0 = \lim_{x' \rightarrow x_0} f(x') - f(x)$$

$$= \lim_{x' \rightarrow x} \frac{f(x') - f(x)}{x' - x} (x' - x)$$

$$= f'(x) \cdot 0$$

$$= 0. \quad \square$$

Rules for differentiation :

1) If f & g are diff'ble
then $f+g$ is and

$$(f+g)' = f' + g'.$$

2) If f & g are diff'able
then $f * g$ is and

$$(f * g)' = f' * g + f * g'$$

↳ this will be on the exc.
sheet.

(Hint: add zero)

Ex. Knowing these rules allows
us to compute the

derivative of all polynomial
functions.

$$f(x) = \sum_{k=0}^{\infty} a_k x^k \quad \text{for } a_k \in \mathbb{R}.$$

then $f'(x) = \sum_{k=0}^{n-1} a_{k+1} \cdot (k+1) x^k.$

Defn x_0 is a local max of f if
 $\exists \varepsilon > 0$ st. $\forall x: |x - x_0| < \varepsilon: f(x) \leq f(x_0)$

Prop. let $f: \mathbb{R} \rightarrow \mathbb{R}$ be
 a function. If f is differentiable at x_0

and $x_0 \in \mathbb{R}$ is a local
 max/min. then $f'(x_0) = 0$.

Pf. It is enough to show

this for maximizer.

1) If x_0 is a local min for f , then it is a max for $(-1)f$.

2) $(-1)f$ is diff'ble

$$3) (-f)'(x_0) = 0 \iff$$

$$f'(x_0) = 0$$

Pf. We saw that limit

preserve inequalities!

$$f(x') - f(x_0)$$

We know $f'(x_0) = \lim_{x' \rightarrow x_0} \frac{f(x') - f(x_0)}{x' - x_0}$

Thus for any sequence (w/ $x_n \neq x_0$)
 $(y_n)_n$ with $y_n \rightarrow x_0$

we have $f'(x_0) = \lim_{n \rightarrow \infty} \frac{f(y_n) - f(x_0)}{y_n - x_0}$

let $(y_n^u)_n$ be seq. of real numbers
with $x_0 < y_n^u$ w/ $y_n^u \rightarrow x_0$

$$\text{then } \left(\frac{f(y_n^u) - f(x_0)}{y_n^u - x_0} \right) < 0.$$

hence it follows.

$$\lim_{n \rightarrow \infty} \left(\frac{f(a) - f(x_0)}{a - x_0} \right) = f'(x_0) \leq 0.$$

Similarly. we can choose $(y_n^l)_n$

with $y_n^l < x_0$ for

and $y_n^l \rightarrow x_0$

therefore
$$\lim_{n \rightarrow \infty} \frac{f(y_n^l) - f(x_0)}{y_n^l - x_0} = f'(x_0) \geq 0,$$

hence and
$$0 \leq f'(x_0) \leq 0.$$



Rem. f' is not necessarily
continuous!

Picture



Application:

Theorem, let f be diff'ble.

Then for any interval $[a, b]$
($a < b$) there exists

$$x_0 \in (a, b)$$

$$f'(x_0) = \frac{f(b) - f(a)}{b - a}$$

Assume in the same situation

but we assume that $f(a) = f(b)$

Then the theorem holds.

$$(f'(x_0) = 0)$$

Pf. We have seen

$f'(x_0)$ vanishes at local
maxima / minima!

By cases, if f is constant,

then $f' = 0$ so we're ~~fine~~!

• If f is not const.

then $\exists x' \in (a, b)$ s.t.

$$f(x) \neq f(a) \quad \circ$$

Fact, A cont function admits
a minimum & maximum on
every closed interval $[a, b]$

So there exists $x_{\max}, x_{\min}$
 $\in [a, b]$ s.t. $f(x_{\max})$
is a max

$$\Delta f(x_{\min})$$

a min.

We need to exclude the
end points!

$$1) \text{ If } f(x) > f(a)$$

$$\text{thus } x_{\max} \in (a, b),$$

$$2) \text{ Conversely if } f(x) < f(a)$$

$$\text{then } x_{\min} \in (a, b).$$

In case 1) $x_0 = x_{\max}$.

2) $x_0 = x_{\min}$.

Then we're done by

the previous prop. 

Pf of theorem:

For the general case:

we look at the following

function

$$\tilde{f}(x) = f(x) + \underbrace{(x-a)}_{\text{}} \left[\frac{f(b) + f(a)}{b-a} \right]$$

$$\bullet \tilde{f}(a) = f(a) = \tilde{f}(b)$$

$$\bullet \tilde{f}'(x) = f'(x) + 1 \cdot \frac{f(b) + f(a)}{b-a}$$

$$\text{So } \tilde{f}'(x_0) = 0$$

$$\Rightarrow f'(x) = \frac{f(b) - f(a)}{b-a}$$



What we did so far:

We defined several notions
of convergence.

→ six instances of convergence.

→ we would have to
improve them every time.

In the library every thing
is done using filters.

They unify all possible aspects
of convergence / continuity.